

Jasper Sands

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RESEARCH INTERESTS

Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for qubit characterization and control.

EDUCATION

Columbia University, New York, NY Sep 2025 – May 2027

Master of Science in Computer Science, Quantum Computing Track. GPA 3.95/4.0.

Thesis: *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

Coursework: Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning.

Washington University in St. Louis, St. Louis, MO Aug 2021 – May 2025

Bachelor of Science in Computer Science and Economics; Minor in Political Science. GPA 3.81/4.0.

TECHNICAL REPORTS

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* Columbia University, 2025. jaspersands.com/assets/practical-llm-based-lossless-compression.pdf

RESEARCH EXPERIENCE

Diraq and UNSW Sydney · *MS Thesis Research* Sep 2026 – May 2027

- Automating silicon spin qubit characterization under Dr. Nard Dumoulin Stuyck, targeting the manual tuning and interpretation of charge measurements that currently bottleneck semiconductor quantum hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing infrastructure and classify charge transitions and device states with machine learning.
- Evaluating the automated analysis against how an expert reads the same device by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and tuning for precision against a fixed false-positive budget.

Desdr Open Insurance Toolkit, Columbia University · *Graduate Researcher* Sep 2025 – Dec 2025

- Built climate-risk and actuarial features from SMS survey data under severe sparsity with Dr. Eugene Wu, modeling expected agricultural losses to support micro-insurance program design.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian by grounding outputs in ontology and molecular graph data.

Foster System Database, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* MIT iQuHACK 2026

- Wrote the GPU-accelerated state preparation, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation. Measured query scaling matched the $O(1/\epsilon)$ bound against an $O(1/\epsilon^2)$ classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* qcompiler.jaspersands.com

- Symplectic stabilizer simulator for rotated and XZZX surface codes, decoding syndromes client-side with exact MWPM and Union-Find. Finite-size scaling estimates thresholds of 3.2% with MWPM and 2.7% with Union-Find.

- Q-Search:** proof-gated quantum algorithm research and verification engine qsearch.jaspersands.com
- Automated search for structural quantum advantage on non-abelian hidden subgroup, dihedral coset, and linear code equivalence problems, with formal proof gates testing each proposed mechanism against 1,200+ classical baselines.
- Noise-Aware Adiabatic Simulation:** transverse-field Ising model · *Qiskit* IBM Heron hardware
- Digitized adiabatic ground-state preparation on Heron's native fractional RZZ gates, roughly halving logical-to-physical translation error, with palindromic Trotter steps for second-order accuracy.

TEACHING EXPERIENCE

- Introduction to Quantum Computing**, Columbia University · *TA* Jan 2026 – May 2026
- Parallel and Concurrent Programming**, Washington University in St. Louis · *TA* Jan 2025 – May 2025
- Reverse Engineering and Malware Analysis**, Washington University in St. Louis · *TA* Aug 2024 – Dec 2024

INDUSTRY EXPERIENCE

- Smack Technologies** · *AI Intern* May 2026 – Aug 2026
- Built physics and sensor simulation for the core pipelines, integrating the knowledge graph to expand corpus generation.
 - Benchmarked open-weight models across task performance, inference speed, and concurrency, increasing corpus-generation throughput 64×.
 - Validated simulator outputs against independently selected reference cases, bounding error to 0.55%, in an automated report that regenerates as the physics model changes.
- Highnote** · *Cybersecurity Engineer Intern* May 2024 – Jul 2024
- Designed and deployed the company-wide SIEM for a card-issuing fintech, ingesting over 1 TB/day.
- Mindtrip** · *Software Engineer Intern* May 2023 – Jul 2023
- Built internal admin and secure-query tooling through Retool, saving roughly 250 engineering hours a month.
- Bugcrowd** · *Associate Application Security Engineer* May 2020 – Jul 2020
- Triaged and validated vulnerability submissions from a global researcher community for enterprise clients.

AWARDS AND COMPETITIONS

- Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street and Classiq challenge.
- Qualified**, NYU Abu Dhabi Quantum Hackathon for Social Good 2027.
- Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026.

TECHNICAL SKILLS

Quantum software: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP · statevector and noise-aware simulation, transpilation, backend execution and benchmarking.

Quantum methods: Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT and block encoding, linear combination of unitaries, quantum walks, VQE and QAOA, Trotter decomposition.

Error correction: Rotated and XZZX surface codes, stabilizer formalism, minimum-weight perfect matching and Union-Find decoding, threshold estimation, randomized benchmarking, tomography, noise modeling.

Programming and tools: Python, C++, Rust, C, MATLAB, JavaScript, Go, Java · PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy · Linux, Git, Docker, WebAssembly, AWS, GCP, LaTeX.

LEADERSHIP AND SERVICE

- Graduate Lead**, Columbia Quantum Algorithms Reading Group (Sep 2025 – Jun 2026). Ran weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms lecture notes, each paired with simulation checks.
- Founding-semester member**, Columbia Autonomous Racing Club. Coordinate sponsor outreach and track testing.